



Acids, alkalis and salts

Name:

Date:

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All acids, when dissolved in water, react in similar ways with metals, metal oxides and metal carbonates. Why do acids have these similar properties?

A Acids are compounds with hydrogen in their chemical formula.

B Acids dissolve in water to produce a solution containing ions.

C Hydrogen atoms in an acid can be replaced by metal atoms to form a salt.

D Acids produce hydrogen ions when they dissolve in water.

Correct Answer: A B C D

Explanation:

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A student tested two solutions and recorded these results:

- pH = 1 for dilute hydrochloric acid.
- pH = 3 for a sample of vinegar.

How much lower was the concentration of hydrogen ions in vinegar compared to that of dilute hydrochloric acid?

A 2 times lower

B 3 times lower

C 100 times lower

D 1000 times lower

Correct Answer: A B C D

Explanation:

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The equations show what happens when four compounds dissolve in water. Which one of the named compounds is not an alkali?

A Ammonia: $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$

B Barium hydroxide: $\text{Ba}(\text{OH})_2(\text{s}) + \text{water} \rightarrow \text{Ba}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$

C Hydrogen sulfide: $\text{H}_2\text{S}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{HS}^-(\text{aq})$

D Methylamine: $\text{CH}_3\text{NH}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CH}_3\text{NH}_3^+(\text{aq}) + \text{OH}^-(\text{aq})$

Correct Answer: A B C D

Explanation:

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Strontium is in group 2 of the periodic table. What is the formula of the salt formed when nitric acid is neutralised by strontium hydroxide?

A SrNO_3 B $\text{Sr}(\text{NO}_3)_2$
C SrNO_2 D $\text{Sr}(\text{NO}_2)_2$

Correct Answer: A B C D

Explanation:

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